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INORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Quantum Number & Electronic Configuration

(English Medium)

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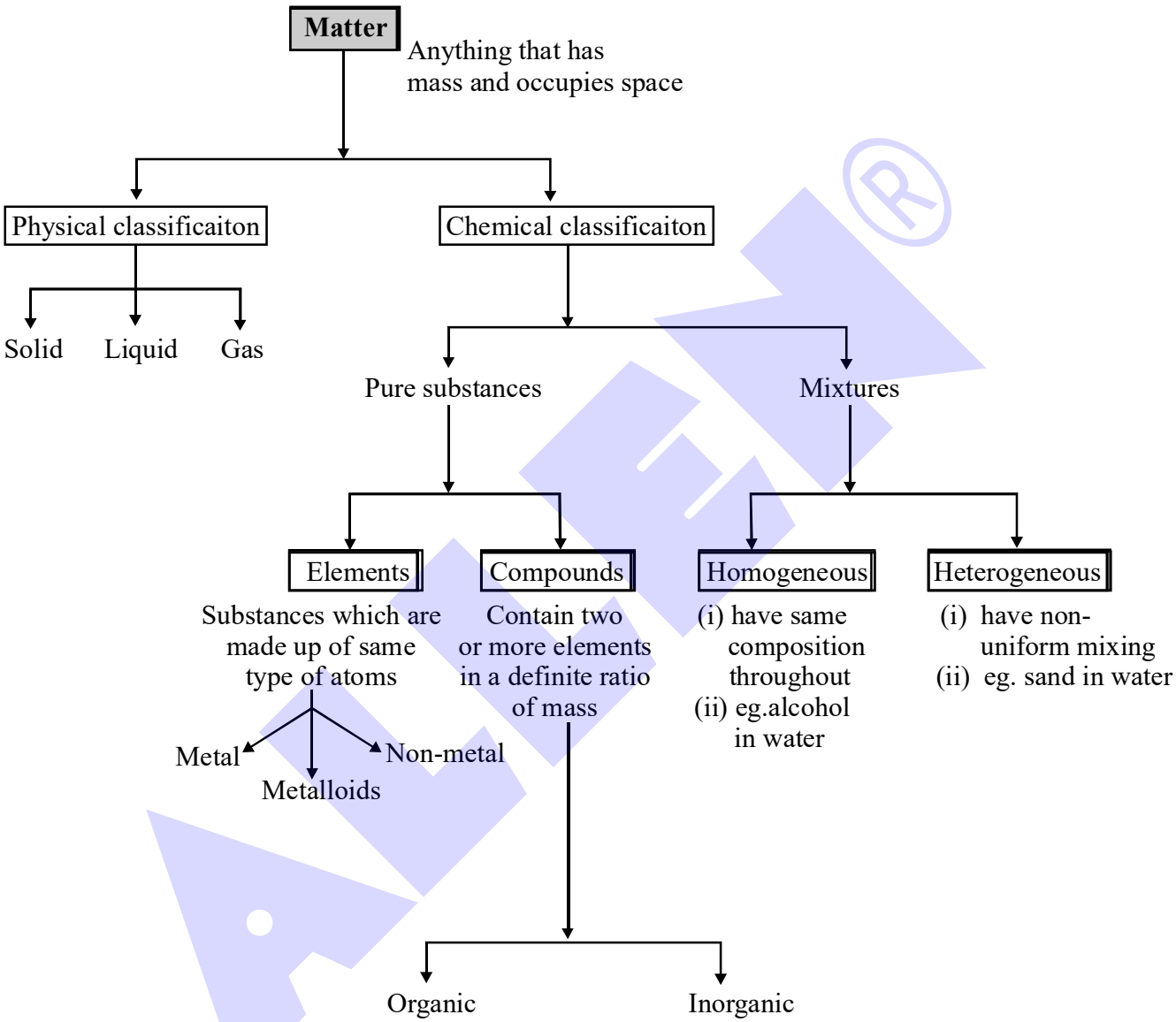
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QUANTUM NUMBER & ELECTRONIC CONFIGURATION

MATTER & ITS CLASSIFICATION



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Example.1 Which of the following is homogeneous mixture :

(A) Oil + Water (B) Milk

(C) Salt dissolved in water (D) All of these

Example.2 Which of the following molecule is tetra-atomic :

(A) CH_2Cl_2 (B) NH_3 (C) H_2O (D) Both (B) and (C)

ATOM : An atom is the smallest particle of an element (made up of still smaller particle like electrons, protons, neutrons etc.) which can take part in a chemical reaction. It may or may not exist free in nature.

Name of particle	Mass	Nature of charge	Amount of charge	Presence in the atom
(i) Electron symbol = (e) Notation = $_{-1}e^{\circ}$ Discoverer J.J. Thomson (1897)	9.11×10^{-28} g $\frac{1}{1837}$th of H-atom	Negatively charged	-1.602×10^{-19} Coulomb or -4.8×10^{-10} e.s.u	Outside the nucleus
(ii) Proton symbol = (p) Notation = ($_1H^1$) Discoverer Rutherford (1911)	1.6725×10^{-24}g	Positively charged	$+ 1.602 \times 10^{-19}$ coulomb $+ 4.8 \times 10^{-10}$ e.s.u.	Inside the nucleus of an atom
(iii) Neutron symbol = (n) Notation = ($_0n^1$) Discoverer J. Chadwick (1932)	1.675×10^{-24} g	Neutral	0	Inside the nucleus of an atom

Representation of atom : $_ZX^A$

A → Mass number : (total number of protons + total number of neutrons present in an atom.)

Z → Atomic number : (total number of protons present in an atom.)

⇒ ***Isotope :*** Atoms of given element which have same atomic number but different mass number are called isotope : e.g. $_1H^1$, $_1H^2$, $_1H^3$ etc.

⇒ ***Isobar :*** Atoms of different elements with the same mass number but different atomic number . e.g. $_{18}Ar^{40}$, $_{19}K^{40}$ and $_{20}Ca^{40}$

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⇒ ***Iso-electronic species :*** Species (atom, molecules or ions) having same number of electrons are called iso-electronic e.g H^-, He, Li^+ and Be^{2+} have 2 valence electrons each.

Note : Now a days this concept is extended to consider the same valence shell electron also.

⇒ ***Iso-sters :*** Species having same number of electrons & same number of atoms. eg. N_2O, CO_2

⇒ ***Iso-diaphers :*** Species having same difference in number of neutrons and protons or same number of excess of neutron. eg. $^{19}_{9}F$, $^{23}_{11}Na$

⇒ ***Orbital :*** An orbital is defined as that zone in space where electron is most likely to be found .The orbitals are characterized by a set of 3 quantum numbers (n,*l*,m).

QUANTUM NUMBERS :

Quantum numbers give complete information about an electron or orbital in an atom.

1. Principal Quantum number (n) :

(i) Permissible value of **n → 1 to ∞**

(ii) It represents shell number/energy level

(iii) The energy states corresponding to different principal quantum numbers are denoted by letters K,L,M, N etc.

n	:	1	2	3	4	5	6
Designation of shell	:	K	L	M	N	O	P

(iv) It indicates the distance of an electron from the nucleus.

(v) It also determines the energy of the electron. In general higher the value of 'n', higher is the energy of a electron.

2. Azimuthal Quantum number (l) :

(i) The values of *l* depends upon the value of 'n' and possible values are '0' to (n –1) .

(ii) It gives the name of subshells associated with the energy level and number of subshells within an energy level.

(iii) The different value of '*l*' indicates the shape of orbitals and designated as follows :

Value	**Notation**	**Name**	**Shape**
$l = 0$	s	Sharp	Spherical
$l = 1$	p	Principal	Dumbell
$l = 2$	d	Diffused	Double Dumbell
$l = 3$	f	Fundamental	Complex

(iv) It also determines the energy of orbital along with n.

For a particular energy level/shell energy of subshell is in the following order → **s < p < d < f.**

(v) It gives the total number of orbitals in a subshell & that equals to **(2*l* + 1)** and number of electron in a subshell = **2 (2*l* + 1)**

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3. Magnetic Quantum number (m or m_l) :

(i) The value of m depends upon the value of *l* and it may have integral value $-l$ to $+l$ including zero.

(ii) It gives the number of orbitals in a given subshell and orientation of different orbitals in space.

e.g. for n = 4, *l* = 0 to 3.

l	0	1	2	3
m	0	+1, 0, – 1	+2, 1, 0, –1, –2	+3, +2, +1, 0, –1, –2, –3
Possible Orientation	1	3	5	7
Orbitals	s	p_x , p_y, p_z	$d_{z^2}, d_{x^2-y^2}$ d_{xy}, d_{yz}, d_{xz}	Not in syllabus

(iii) The orbitals having same value of n and *l* but different value of m, have same energy in absence of external electric & magnetic field. These orbitals having same energy of a particular subshell is known as Degenerate orbitals.

4. Spin Quantum number (s) OR magnetic spin quantum number (m_s) :

(i) While moving around the nucleus, the electron always spins about its own axis either clockwise or anticlockwise. The magnetic spin quantum number represents the direction of electron spin (rotation) around its own axis (clockwise or anticlockwise).

(ii) There are two possible values of $\mathbf{m_s}$ are $+\frac{1}{2}$ & $-\frac{1}{2}$ and represented by the two arrows ↑ (spin up) and ↓ (spin down).

Shell No.	**No. of Subshells**	**No. of Orbitals**	**No. of e⁻**
n = 1 (K)	1(1s)	1	2
n = 2 (L)	2(2s,2p)	1 + 3 = 4	8
n = 3 (M)	3(3s,3p,3d)	1 + 3 + 5 = 9	18
n = 4 (N)	4(4s,4p,4d,4f)	1 + 3 + 5 + 7 = 16	32
n = 5 (O)	1(1s)	1 + 3 + 5 + 7 + 9 = 25	50
⋮	⋮	⋮	⋮
n = n	n	n^2	$2n^2$

Sub-Shell No.	**No. of Orbitals**	**No. of electron**
$\ell = 0 \rightarrow$ s	1	2
$\ell = 1 \rightarrow$ p	3	6
$\ell = 2 \rightarrow$ d	5	10
$\ell = 3 \rightarrow$ f	7	14
$\ell = 4 \rightarrow$ g	9	18
⋮	⋮	⋮
$\ell = \ell$	$2\ell + 1$	$2(2\ell + 1)$

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RULES FOR FILLING ELECTRONS :

1. Pauli's exclusion principle

'No two electrons in an atom can have same values of all the four quantum numbers.

An orbital accommodates two electron with opposite spin. These two electrons have same values of principal, azimuthal and magnetic quantum number but the fourth, i.e. magnetic spin quantum number will be different. i.e.

For K, shell (n = 1)

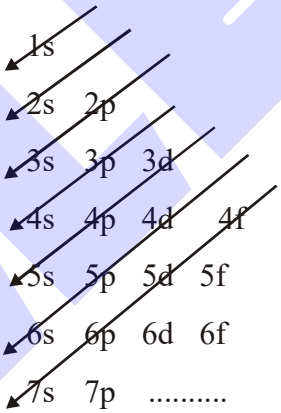
$l = 0$, m = 0

For 1st Electron n = 1, $l = 0$, m = 0, $m_s = +\frac{1}{2}$

For 2nd Electron n = 1, $l = 0$, m = 0, $m_s = -\frac{1}{2}$

2. Aufbau Principle (Means Building up) :

The electrons are added progressively to the various orbitals in the order of increasing energies starting with the orbital of the lowest energy



1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s < 5f < 6d < 7p

Alternatively, the order of increase of energy of orbitals can be calculated from (n + l) rule.

(i) Lower the value of (n + l) for an orbital, the lower will be its energy.

(ii) If two orbitals have the same (n + l) value, then orbital with lower value of n has the lower energy.

e.g. 2p & 3s

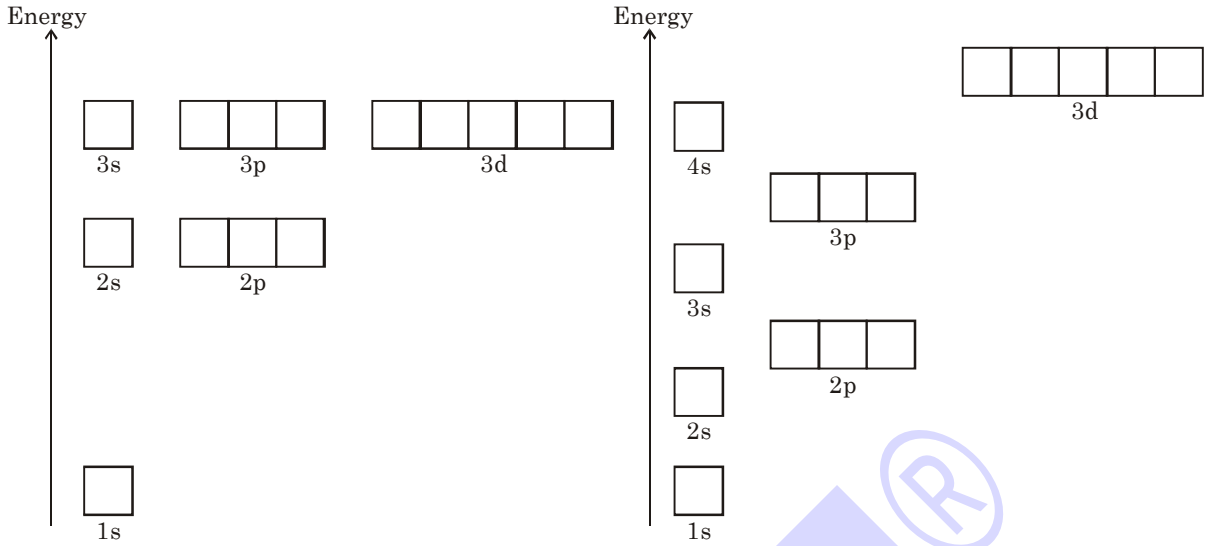
For 2p, n = 2, $l = 1$, (n + l) = 2 + 1 = 3

For 3s, n = 3, $l = 0$, (n + l) = 3 + 0 = 3

Then for 2p, n is lesser than for 3s, so 2p has lower energy than 3s.

(iii) 1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f .... energy order of different orbitals for single electron system like H, He^{+}, Li^{+2} etc.

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(A) For single electron or hydrogenic atom (B) Multi electronic atoms

Energy level diagram for few electronic shells

Example.4 Write the increasing order of energies of 4s, 3p, 4p and 3d.

Ans. For 4s, n = 4, l = 0, (n + l) = 4

For 3p, n = 3, l = 1, (n + l) = 4

For 4p, n = 4, l = 1, (n + l) = 5

For 3d, n = 3, l = 2, (n + l) = 5

$\Rightarrow$ 3p < 4s < 3d < 4p increasing order

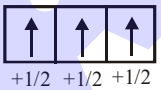
3. Hund's rule of maximum multiplicity :

This rule deals with the filling of electrons into the orbitals belonging to the same subshell i.e. orbitals of equal energy, called degenerate orbitals.

"Electrons are distributed among the orbitals of a subshell in such a way as to give the maximum number of unpaired electron with parallel spins. "

"Pairing of electrons in the orbitals belonging to the same subshell (p, d, f) does not take place until each orbital belonging to that subshell has got one electron each i.e. singly occupied. Moreover, the singly occupied orbitals must have the electrons with the parallel spin multiplicity"

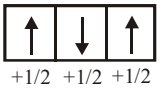
Multiplicity = 2|S| + 1, where S = Total spin.

i.e. 

Find total spin & multiplicity

Total spin $S = \frac{1}{2}+\frac{1}{2}+\frac{1}{2}=\frac{3}{2}$

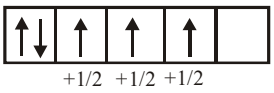
Multiplicity $= 2\times\frac{3}{2}+1=4$

i.e. 

Total spin $S = \frac{1}{2}-\frac{1}{2}+\frac{1}{2}=\frac{1}{2}$

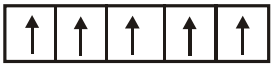
Multiplicity $= 2\times\frac{1}{2}+1=2$

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i.e. 

Total spin S = $\frac{1}{2}+\frac{1}{2}+\frac{1}{2}=\frac{3}{2}$

Multiplicity = $2\times\frac{3}{2}+1=4$

i.e. 

Total spin = $5\times\frac{1}{2}=\frac{5}{2}$

Multiplicity = $2\times\frac{5}{2}+1=6$

ELECTRONIC CONFIGURATION OF ATOMS :

The distribution of electrons in various shells, subshells and orbitals, in an atom of an element, is called its electronic configuration.

x ← no. of electrons in subshell

n*l*

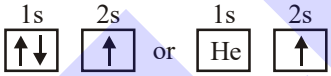
└ represents subshell

(represents shell)

$2p^5$ ← no. of electrons in p-subshell

└ p subshell

Second shell

$_3Li$; $1s^2, 2s^1$ or [He] $2s^1$ or 

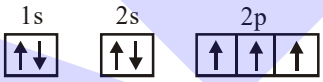
Electronic configuration :

Example.5 Nitrogen

Ans. $_7N$: $1s^2, 2s^2\ 2p^3$ [Orbital notation method]

$1s^2, 2s^2\ 2p_x^{\ 1}\ 2p_y^{\ 1} 2p_z^{\ 1}$

or



[Orbital diagram method]

or

[He] $2s^2\ 2p^3$ [Condensed form]

Extra stability of Half-filled and fully-filled orbitals.

The electronic configuration of most of the atoms follows the Aufbau's rule. However, in certain elements such as Cr, Cu etc. Where the two subshells (4s and3d) differ slightly in their energies (4s < 3d), an electron shifts from a subshell of lower energy (4s) to a subshell of higher energy (3d), provided such a shift results in all orbitals of the subshell of higher energy getting either completely filled or half-filled.

$_{24}Cr \rightarrow$ [Ar] $3d^5, 4s^1$ and not [Ar] $3d^4\ 4s^2$

$_{29}Cu \rightarrow$ [Ar] $3d^{10}, 4s^1$ and not [Ar] $3d^9\ 4s^2$

It has been found that there is extra stability associated with these electronic configuration. This stabilization is due to the following two factors.

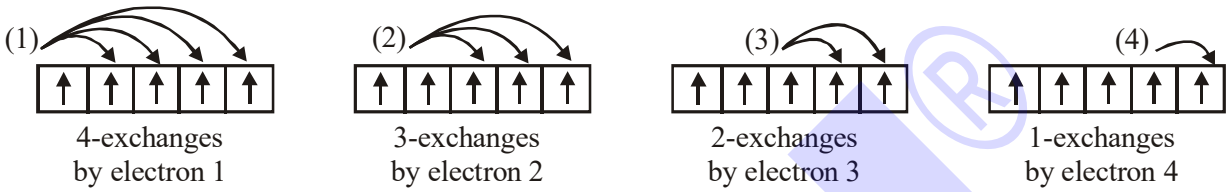
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(i) Symmetrical distribution of electron :

It is well known that symmetry leads to stability. The completely filled or half-filled subshell have symmetrical distribution of electron in them and are therefore more stable. This effect is more dominant in d and f-orbitals. This means three or six electrons in p-subshell, 5 or 10 electrons in d-subshell and 7 or 14 in f-subshell forms a stable arrangement.

(ii) Exchange energy :

This stabilizing effect arises whenever two or more electrons with the same spin are present in the degenerate orbitals of a subshell. these electrons tend to exchange their positions and the energy released due to this exchange is called exchange energy. The number of exchanges that can take place is maximum when the subshell is either half filled or fully filled. As result the exchange energy is maximum and so is the stability.

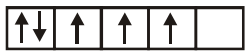


4-exchanges by electron 1 | 3-exchanges by electron 2 | 2-exchanges by electron 3 | 1-exchanges by electron 4

Total exchange pairs = 10

$\frac{n(n-1)}{2}$ → Number of exchange pairs

n → Number of electron with parallel spins.

e.g. 

Only 6 total exchange possible

Exceptional electronic configuration

S.No.	Element	Z	Configuration
1	Cr	24	$[Ar]4s^1 3d^5$
2.	Cu	29	$[Ar]4s^1 3d^{10}$
3.	Nb	41	$[Kr]5s^1 4d^4$
4.	Mo	42	$[Kr]5s^1 4d^5$
5.	Ru	44	$[Kr]5s^1 4d^7$
6.	Rh	45	$[Kr]5s^1 4d^8$
7.	Pd	46	$[Kr]4d^{10}$
8.	Ag	47	$[Kr]5s^1 4d^{10}$
9.	La	57	$[Xe]6s^2 5d^1$
10.	Pt	78	$[Xe]6s^1 4f^{14} 5d^9$
11.	Au	79	$[Xe]6s^1 4f^{14} 5d^{10}$
12.	Ac	89	$[Rn]7s^2 6d^1$
13.	Th	90	$[Rn]7s^2 6d^2$

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MAGNETIC PROPERTIES :

❖ ***Paramagnetism :***

(i) The substances which are weakly attracted by magnetic field are paramagnetic and this phenomenon is known as paramagnetism.

(ii) Generally species which have unpaired electrons are paramagnetic in nature.

❖ ***Diamagnetism :***

(i) The substances which are weakly repelled by magnetic field are diamagnetic and this phenomenon is known as diamagnetism.

(ii) Diamagnetic substances lack unpaired electrons and their magnetic moment is zero.

❖ ***Spin magnetic moment :***

The spin magnetic moment of electron (excluding orbit magnetic moment) is given by :

$\mu = \sqrt{[n(n+2)]}$ B.M.

Where n is number of unpaired electron in species.

The magnetic moment is expressed in Bohr magneton (B.M.)

Example.6 A compound of vanadium has magnetic moment of 1.73 BM. Work out the electronic configuration of vanadium ion in the compound.

Ans. Vanadium belongs to 3d series with Z = 23. The magnetic moment of 3d series metal is given by spin only formula.

$\mu = \sqrt{n(n+2)}$ BM (BM = Bohr's magneton)

$\because$ $1.73 = \sqrt{3}$

$\Rightarrow$ $n(n+2) = 3 \Rightarrow n = 1$

$\Rightarrow$ Magnetic moment correspond to one unpaired electron.

$\Rightarrow$ Electronic configuration of vanadium atom $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^3$.

For one unpaired electron 4 electron must be removed in which first 2 electron are lost from 4s orbital (outermost).

Electronic configuration of V^{+4}

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^1$

Nodal Planes of different orbitals :

Nodal plane is a plane at which the probability of finding an electron becomes zero.

Eg.

Orbital	Nodal plane	Orbital	Nodal plane
s	None	d_{xy}	XZ & YZ planes
p_x	YZ plane	d_{yz}	XZ & XY planes
p_y	XZ plane	d_{xz}	XY & YZ planes
p_z	XY plane	$d_{x^2-y^2}$	Planes perpendicular to XY plane, passing through origin (nucleus) and inclined at 45° to X & Y axis.
		d_{z^2}	None (two nodal cones are available)

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SCREENING EFFECT (s) AND EFFECTIVE NUCLEAR CHARGE (Z_{eff})

(a) Valence shell electron suffer force of attraction due to nucleus and force of repulsion due to inner shell electrons.

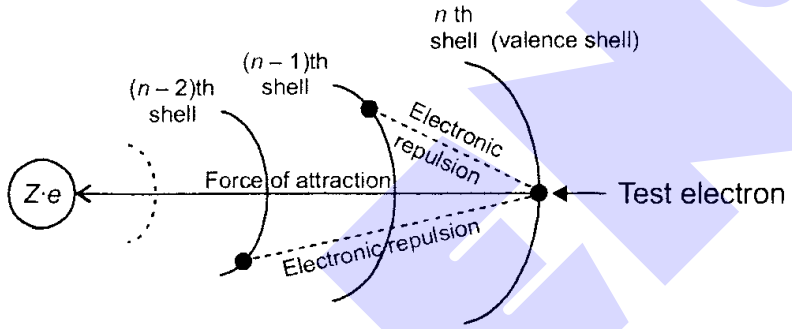
(b) The decrease in force of attraction on valence electron due to inner shell electron is called screening effect or shielding effect.(i.e. total repulsive force is called shielding effect.)

(c) Due to screening effect, valence shell electron experiences less force of attraction by nucleus.

(d) Due to screening effect, net attractive force felt by the electron is measured by effective nuclear charge, Z_{eff}

(e) If nuclear charge = Z, then effective nuclear charge = $Z - \sigma$ (Where σ 'sigma' is called screening constant/sheilding constant)

So, $Z_{eff} = Z - \sigma$



CALCULATION OF σ (using slater's rule)

To calculate the shielding constant (σ) :

(a) Write the electronic configuration of the element in the following order and groupings :
(1s), (2s, 2p), (3s, 3p), (3d), (4s, 4p), (4d), (4f), (5s, 5p), etc.

For s and p electrons :

(b) Electrons in any group to the right of the (ns, np) group contribute nothing to the shielding constant.
(n-shell no. of the electron for which σ is calculated)

(c) All of the other electrons in the (ns, np) group, shield the concern electron to an extent of 0.35 each. (Except for the 1s orbital for which value is 0.30).

(d) All electrons in the (n – 1) shell shield to an extent of 0.85 each.

(e) All electrons (n – 2) or lower group shield completely ; that is, their contribution is 1.00 each.

For d and f electrons :

(f) Electrons in any group to the right of the nd or nf group contribute nothing to the shielding constant.

(g) All the other electrons in the nd or nf group, shield the valence electron to an extent of 0.35 each.

(h) All electrons in groups lying to the left of the nd or nf group contribute 1.00.

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(Effective Nuclear charge of elements of second period)						
Element	**Electronic Configaration**	**Z**	**σ of ns & np electron (a)**	**σ (n–1) orbital (b)**	**Total Screeing Constant (a + b)**	**Effective nuclear charge# Z*=Z – σ**
$_3$Li	$1s^2\ 2s^1$	3	–	0.85× 2=1.70	1.70	1.30
$_4$Be	$1s^2,\ 2s^2$	4	1× 0.35=0.35	0.85× 2=1.70	2.05	1.95
$_5$B	$1s^2,2s^2,2p^1$	5	2× 0.35=0.70	0.85× 2=1.70	2.40	2.60
$_6$C	$1s^2,2s^2,2p^2$	6	3× 0.35=1.05	0.85× 2=1.70	2.75	3.25
$_7$N	$1s^2,2s^2,2p^3$	7	4× 0.35=1.40	0.85× 2=1.70	3.10	3.90
$_8$O	$1s^2,2s^2,2p^4$	8	5× 0.35=1.75	0.85× 2=1.70	3.45	4.55
$_9$F	$1s^2,2s^2,2p^5$	9	6× 0.35=2.10	0.85× 2=1.70	3.80	5.20

Calculated for valence electron.

Key Points :

(a) From left to right in a period Z_{eff} increases

(i) For s and p-block elements, Z_{eff} in a period increases by 0.65 where atomic number increases by 1, and hence atomic size decreases considerably.

(ii) In transition series Z increase by + 1 but screening constant increases by 0.85 So Z_{eff} is increased by 0.15
(1– 0.85 = 0.15) [Because e^- enters in (n – 1) orbit which has value of σ = 0.85]

(b) From top to bottom in a group Z_{eff} remain constant for s-block elements, after Li and Be.

Element	Li	Na	K	Rb	Cs	Fr
Z_{eff}	1.30	2.20	2.20	2.20	2.20	2.20

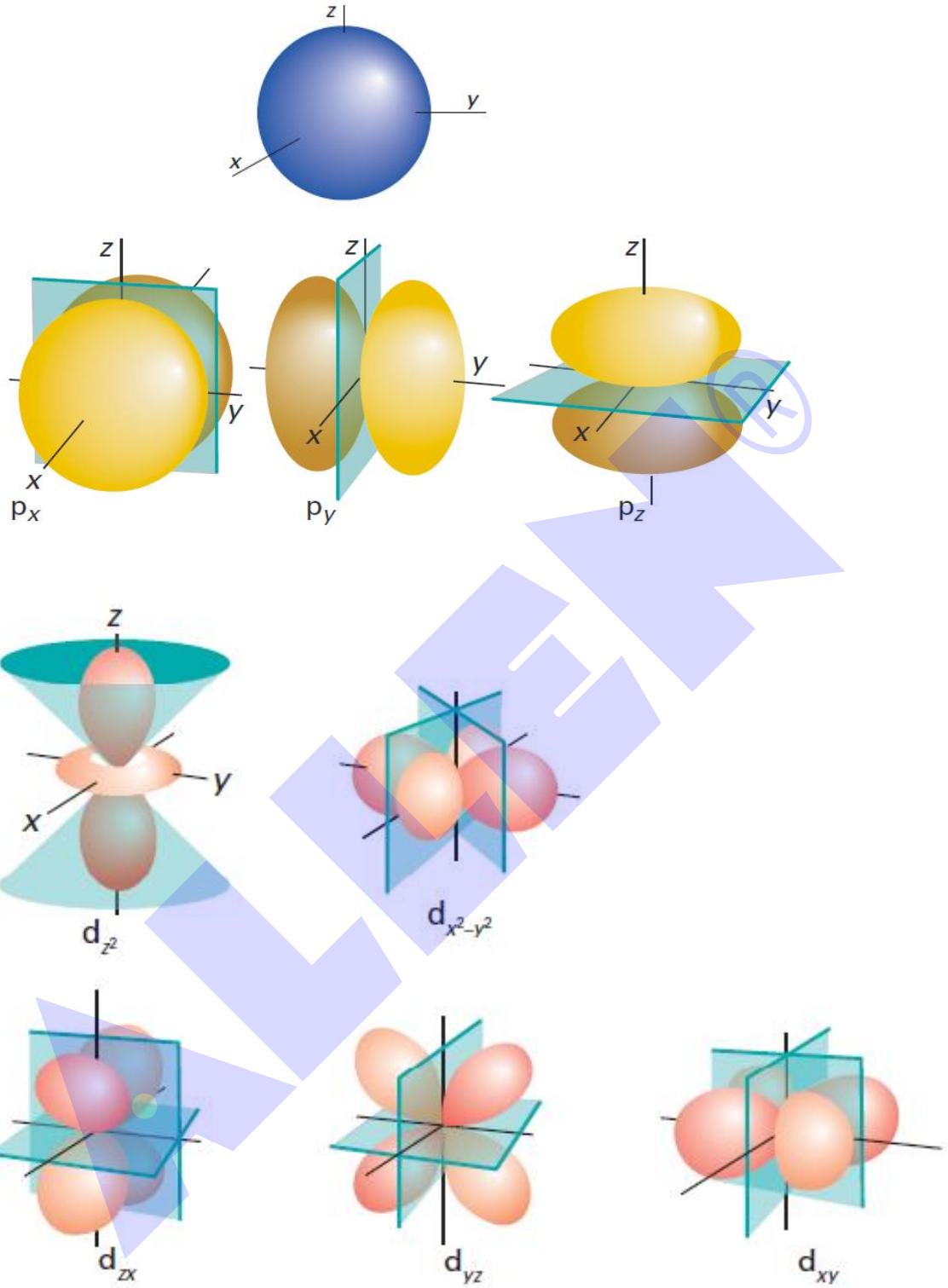
Example-7 :

What is the effective nuclear charge at the periphery of nitrogen atom when an extra electron is added during the formation of an anion. Also find the value of Z_{eff} when the atom is ionized to N^+.

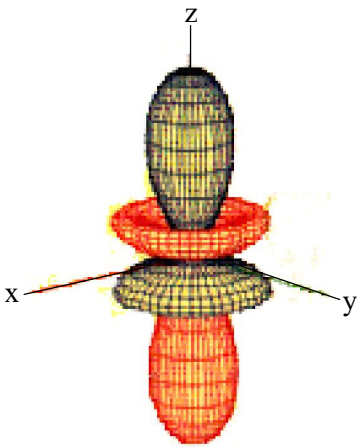
Ans. Ground state electron configuration of N(Z = 7) = $1s^2\ 2s^2\ 2p^3$
Electron configuration of N^- = $(1s^2)\ (2s^2\ 2p^4)$
Shielding constant for the last 2p electron,
σ = [(2 × 0.85) + (5 × 0.35)] = 3.45
So Z_{eff} = Z – σ = 7 – 3.45 = 3.55
Electron configuration of N^+ = $(1s^2)\ (2s^2\ 2p^2)$
Shielding constant for the last 2p electron,
σ = [(2 × 0.85) + (3 × 0.35)] = 2.75
So Z_{eff} for last electron on N^+ = 7 – 2.75 = 4.25

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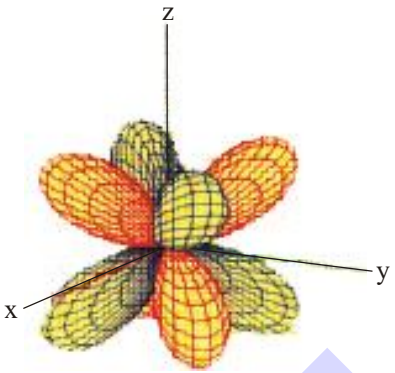
SHAPES OF ATOMIC ORBITALS



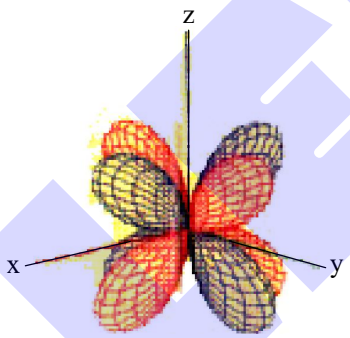
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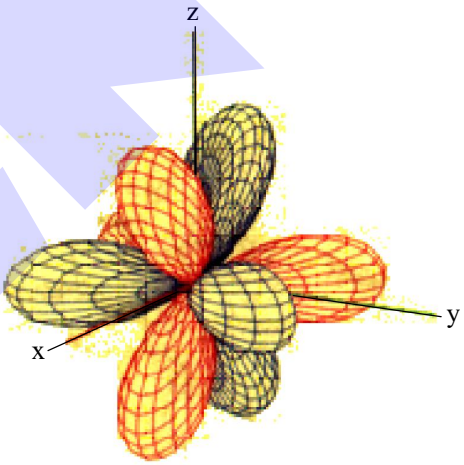
f_{z^3}



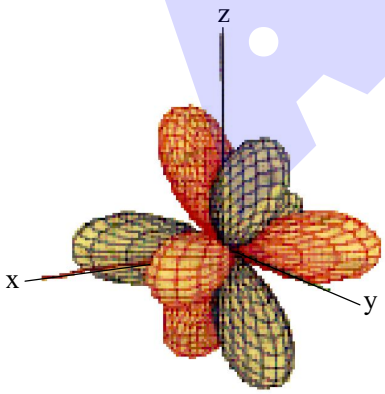
f_{xyz}



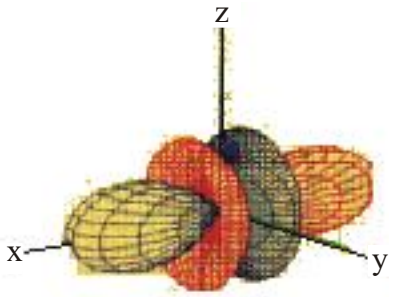
$f_{z(x^2-y^2)}$



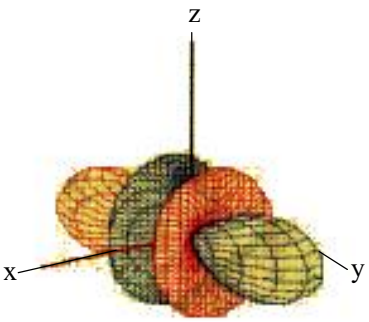
$f_{x(y^2-z^2)}$



$f_{y(z^2-x^2)}$



f_{x^3}



f_{y^3}

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